Unrestricted Coupled Cluster Singles and Doubles for Complex Orbitals

1 Scope

The standard spin-orbital CCSD equations (Stanton, Gauss, Watts, Bartlett, J. Chem. Phys. **94**, 4334 (1991)) are written for real orbitals. In that form several integrals appear in a particular index order that is only correct because of the real-orbital symmetry $\langle pq\|rs\rangle = \langle rs\|pq\rangle$. Here the equations are re-derived by tracking which operator string every integral multiplies, so that they hold for *arbitrary complex* spin-orbitals $\varphi_p(\mathbf{r},\sigma)$. This covers

- UCCSD with complex spatial orbitals, $\varphi_p = \psi_p(\mathbf{r})\,\sigma_p$ with $\sigma_p \in \{\alpha,\beta\}$ (Section 7);
- general two-component or four-component spinors (the case needed for the relativistic code), where there is no spin factorization at all.

No time-reversal or other symmetry is assumed. The reference determinant $|0\rangle$ is built from the N occupied spin-orbitals and need not be a Hartree–Fock solution (the Fock matrix may have non-zero occupied–virtual blocks).

2 Conventions

Indices i,j,k,l,m,n label occupied, a,b,c,d,e,f virtual, p,q,r,s general spin-orbitals. The one-electron operator is h, with $h_{pq} = \langle p|h|q\rangle = \int \varphi_p^*\, h\, \varphi_q$. Integrals:

$$\langle pq|rs\rangle = \iint \varphi_p^*(1)\,\varphi_q^*(2)\,\frac{1}{r_{12}}\,\varphi_r(1)\,\varphi_s(2), \tag{1}$$

$$\langle pq\|rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle. \tag{2}$$

The Hamiltonian in second quantization,

$$H = \sum_{pq} h_{pq}\, a_p^\dagger a_q + \tfrac{1}{4}\sum_{pqrs}\langle pq\|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r, \tag{3}$$

so that $\langle pq\|rs\rangle$ is always the coefficient of the string $a_p^\dagger a_q^\dagger a_s a_r$: *the two left indices are the created orbitals, the two right indices the annihilated ones.* This single rule fixes every conjugation below.

The Fock matrix and the reference energy are

$$f_{pq} = h_{pq} + \sum_i \langle pi\|qi\rangle, \qquad E_0 = \sum_i h_{ii} + \tfrac{1}{2}\sum_{ij}\langle ij\|ij\rangle, \tag{4}$$

$f_{pq} = f_{qp}^*$ (Hermitian, in general complex).

Integral symmetries

	real orbitals	complex orbitals
$\langle pq\|\|rs\rangle = -\langle qp\|\|rs\rangle = -\langle pq\|\|sr\rangle = \langle qp\|\|sr\rangle$	yes	yes
$\langle pq\|\|rs\rangle = \langle rs\|\|pq\rangle$	yes	**no**
$\langle pq\|\|rs\rangle = \langle rs\|\|pq\rangle^*$	(trivial)	yes
$f_{pq} = f_{qp}$	yes	**no**, $f_{pq} = f_{qp}^*$

The 8-fold permutational symmetry of the real case is reduced to 4-fold for the unantisymmetrized $\langle pq|rs\rangle$ (namely $\langle pq|rs\rangle = \langle qp|sr\rangle = \langle rs|pq\rangle^* = \langle sr|qp\rangle^*$). In particular, the blocks with virtuals in the bra and occupied in the ket (e.g. $\langle ab\|ij\rangle$) are the complex conjugates of the corresponding blocks with occupied in the bra (e.g. $\langle ij\|ab\rangle$), and are not equal to them.

3 Cluster ansatz and amplitude equations

$$|\Psi\rangle = e^T|0\rangle, \qquad T = T_1 + T_2, \tag{5}$$

$$T_1 = \sum_{ia} t_i^a\, a_a^\dagger a_i, \qquad T_2 = \tfrac{1}{4}\sum_{ijab} t_{ij}^{ab}\, a_a^\dagger a_b^\dagger a_j a_i, \tag{6}$$

with t_{ij}^{ab} antisymmetric in (ij) and in (ab). The amplitudes are complex and satisfy no conjugation relation such as $t_{ij}^{ab} = t_{ji}^{ba*}$. The equations are the usual projections of the similarity-transformed Hamiltonian $\bar{H} = e^{-T}He^T$,

$$E_c = \langle 0|\bar{H}|0\rangle - E_0, \qquad \langle \Phi_i^a|\bar{H}|0\rangle = 0, \qquad \langle \Phi_{ij}^{ab}|\bar{H}|0\rangle = 0. \tag{7}$$

Neither the T amplitudes nor the left-hand projectors are ever complex conjugated. Complex orbitals therefore enter *only* through the integrals, and the algebra is identical to the real case term by term; what changes is which of the two members of a conjugate pair $\langle pq\|rs\rangle$, $\langle rs\|pq\rangle$ appears. There is no left-hand state and no Λ equation in what follows.

4 Working equations

4.1 Correlation energy

$$E_c = \sum_{ia} f_{ia}\, t_i^a + \tfrac{1}{4}\sum_{ijab}\langle ij\|ab\rangle\, t_{ij}^{ab} + \tfrac{1}{2}\sum_{ijab}\langle ij\|ab\rangle\, t_i^a t_j^b. \tag{8}$$

For the exact wave function E_c is real. For truncated CCSD no identity guarantees it, so take $\mathrm{Re}\, E_c$ and monitor $\mathrm{Im}\, E_c$ (PySCF's GCCSD warns when it exceeds 10^{-4}).

4.2 Intermediates

With

$$\tilde{\tau}_{ij}^{ab} = t_{ij}^{ab} + \tfrac{1}{2}\left(t_i^a t_j^b - t_i^b t_j^a\right), \qquad \tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a, \tag{9}$$

and $P(pq)X_{pq} = X_{pq} - X_{qp}$:

$$F_{ae} = (1-\delta_{ae})f_{ae} - \tfrac{1}{2}\sum_m f_{me}\,t^a_m + \sum_{mf} t^f_m \langle ma\|fe\rangle - \tfrac{1}{2}\sum_{mnf} \tilde{\tau}^{af}_{mn}\langle mn\|ef\rangle, \tag{10}$$

$$F_{mi} = (1-\delta_{mi})f_{mi} + \tfrac{1}{2}\sum_e t^e_i f_{me} + \sum_{ne} t^e_n \langle mn\|ie\rangle + \tfrac{1}{2}\sum_{nef} \tilde{\tau}^{ef}_{in}\langle mn\|ef\rangle, \tag{11}$$

$$F_{me} = f_{me} + \sum_{nf} t^f_n \langle mn\|ef\rangle, \tag{12}$$

$$W_{mnij} = \langle mn\|ij\rangle + P(ij)\sum_e t^e_j \langle mn\|ie\rangle + \tfrac{1}{4}\sum_{ef} \tau^{ef}_{ij}\langle mn\|ef\rangle, \tag{13}$$

$$W_{abef} = \langle ab\|ef\rangle - P(ab)\sum_m t^b_m \langle am\|ef\rangle + \tfrac{1}{4}\sum_{mn} \tau^{ab}_{mn}\langle mn\|ef\rangle, \tag{14}$$

$$W_{mbej} = \langle mb\|ej\rangle + \sum_f t^f_j \langle mb\|ef\rangle - \sum_n t^b_n \langle mn\|ej\rangle - \sum_{nf}\left(\tfrac{1}{2}t^{fb}_{jn} + t^f_j t^b_n\right)\langle mn\|ef\rangle. \tag{15}$$

Each of these multiplies, in $\bar{H}$, an operator string whose created indices are the left indices (e.g. F_{ae} multiplies $a^\dagger_a a_e$ and W_{abef} multiplies $a^\dagger_a a^\dagger_b a_f a_e$), and every integral inside them keeps the created-left / annihilated-right pattern of that string. This is why these seven definitions need no conjugate: all integrals appearing there are already in the orientation that the operator algebra delivers.

4.3 T_1 equation

$$t^a_i D^a_i = f_{ai} + \sum_e t^e_i F_{ae} - \sum_m t^a_m F_{mi} + \sum_{me} t^{ae}_{im} F_{me} - \sum_{nf} t^f_n \langle na\|if\rangle \\ - \tfrac{1}{2}\sum_{mef} t^{ef}_{im}\langle ma\|ef\rangle - \tfrac{1}{2}\sum_{men} t^{ae}_{mn}\langle nm\|ei\rangle. \tag{16}$$

4.4 T_2 equation

$$t^{ab}_{ij} D^{ab}_{ij} = \langle ab\|ij\rangle + P(ab)\sum_e t^{ae}_{ij}\Big(F_{be} - \tfrac{1}{2}\sum_m t^b_m F_{me}\Big) - P(ij)\sum_m t^{ab}_{im}\Big(F_{mj} + \tfrac{1}{2}\sum_e t^e_j F_{me}\Big) \\ + \tfrac{1}{2}\sum_{mn}\tau^{ab}_{mn}W_{mnij} + \tfrac{1}{2}\sum_{ef}\tau^{ef}_{ij}W_{abef} + P(ij)P(ab)\sum_{me}\Big(t^{ae}_{im}W_{mbej} - t^e_i t^a_m\langle mb\|ej\rangle\Big) \\ + P(ij)\sum_e t^e_i\langle ab\|ej\rangle - P(ab)\sum_m t^a_m\langle mb\|ij\rangle. \tag{17}$$

The denominators contain the diagonal of the (Hermitian, hence real-diagonal) Fock matrix,

$$D^a_i = f_{ii} - f_{aa}, \qquad D^{ab}_{ij} = f_{ii} + f_{jj} - f_{aa} - f_{bb}. \tag{18}$$

The off-diagonal Fock elements enter through the $(1-\delta)$ terms in F_{ae}, F_{mi}, so no canonical-orbital assumption is made. Level shifting and DIIS act on the complex amplitude vector as in the real case, with the DIIS overlap matrix taken as $\mathbf{e}^\dagger_k\mathbf{e}_l$ and the extrapolation coefficients real.

5 Differences with respect to the real-orbital equations

Compared with the equations as usually printed for real orbitals, only the two inhomogeneous terms change, and they change because the real-orbital formulas are written with the occupied-bra orientation using $\langle pq\|rs\rangle = \langle rs\|pq\rangle$:

term	real-orbital form	complex form (this note)
T_1 inhomogeneous term	f_{ia}	$f_{ai} = f_{ia}^*$
T_2 inhomogeneous term	$\langle ij\|ab\rangle$	$\langle ab\|ij\rangle = \langle ij\|ab\rangle^*$

All other terms, including $P(ij)\, t_i^e \langle ab\|ej\rangle$ and $-P(ab)\, t_m^a \langle mb\|ij\rangle$, are printed with the same index order as in the real case; they are correct for complex orbitals because that order is the created-left / annihilated-right order of the operator algebra. What differs is only how they are obtained from stored arrays: if the code stores the occupied-bra blocks ($\langle ij\|mb\rangle$, $\langle je\|ba\rangle$), then

$$\langle ab\|ej\rangle = \langle je\|ba\rangle^*, \qquad \langle mb\|ij\rangle = \langle ij\|mb\rangle^*, \tag{19}$$

(pair swap $\langle pq\|rs\rangle = \langle qp\|sr\rangle$ combined with Hermitian conjugation) and the conjugate must be taken. For real orbitals every conjugation is trivial and the equations reduce to Stanton's. The same pattern is used in PySCF's `GCCSD` (`pyscf/cc/gccsd.py`, `update_amps`), which can serve as an external reference with complex spin-orbitals.

6 Initial guess

With $f_{ov} = 0$ the first iteration from $T = 0$ gives the MP2 amplitudes

$$t_i^a = \frac{f_{ai}}{D_i^a}, \qquad t_{ij}^{ab} = \frac{\langle ab\|ij\rangle}{D_{ij}^{ab}}, \tag{20}$$

and $E_{\mathrm{MP2}} = \frac{1}{4}\sum \langle ij\|ab\rangle\, t_{ij}^{ab} = \frac{1}{4}\sum |\langle ij\|ab\rangle|^2 / D_{ij}^{ab}$ is real and negative as it should be. Using $\langle ij\|ab\rangle$ instead of $\langle ab\|ij\rangle$ in the numerator would give a complex energy, which is a convenient sanity check on the orientation of the integrals.

7 UCCSD: spin blocks for complex spatial orbitals

For $\varphi_p = \psi_p\, \sigma_p$ with complex spatial ψ_p,

$$\langle pq|rs\rangle = \delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}\,(pr|qs), \qquad (pr|qs) = \iint \psi_p^*(1)\psi_r(1)\,\frac{1}{r_{12}}\,\psi_q^*(2)\psi_s(2), \tag{21}$$

and h, f conserve spin. Spin conservation then restricts the amplitudes to

- t_i^a: $\sigma_i = \sigma_a$, giving t^α and t^β;
- t_{ij}^{ab}: $\sigma_i + \sigma_j = \sigma_a + \sigma_b$, giving the unique blocks $\alpha\alpha \to \alpha\alpha$, $\beta\beta \to \beta\beta$, and $\alpha\beta \to \alpha\beta$ (t_{iJ}^{aB}; the blocks with a, B swapped follow by antisymmetry).

The equations of the previous section hold verbatim in this block structure: the sums run over the spin-allowed combinations and the integral blocks are $\langle pq\|rs\rangle = (pr|qs) - (ps|qr)$ for equal spins (all four orbitals of one spin, or $\sigma_p = \sigma_r, \sigma_q = \sigma_s$) and $\langle pq\|rs\rangle = (pr|qs)$ for the mixed block

$p, r \in \alpha$, $q, s \in \beta$. The conjugation pattern of the previous section applies unchanged to each block; in particular the $\alpha\beta$ inhomogeneous term is $\langle aB\|iJ\rangle = (ai|BJ) = (ia|JB)^*$, with no exchange part.

For genuine spinors (two- or four-component) φ_p is not an eigenfunction of σ_z, the integrals $\langle pq|rs\rangle$ do not factorize, and the spin-orbital equations of Section 4 are used directly with all index combinations, i.e. as a generalized CCSD (GCCSD) with complex orbitals.

8 Cost and storage

The scaling is that of real CCSD, $O(o^2v^4)$ time and $O(o^2v^2)$ amplitude storage, with complex arithmetic (about $4\times$ the flops and $2\times$ the memory per element, using `zgemm` in the contractions). Because $\langle pq\|rs\rangle$ has only 4-fold symmetry, one cannot store only the unique integrals of one orientation and reuse them for the conjugate orientation by a transposition; the conjugate blocks of the previous section are obtained by conjugating the stored block, never by transposing it alone.

9 Left-hand equations (Λ) and reduced density matrices

Everything above is the right-hand side of the CCSD problem. Densities, and any property that is not the energy, need the left-hand state $\langle 0|(1+\Lambda)e^{-T}$,

$$\Lambda = \sum_{ia} \lambda_i^a \, a_i^\dagger a_a + \tfrac{1}{4} \sum_{ijab} \lambda_{ij}^{ab} \, a_i^\dagger a_j^\dagger a_b a_a, \tag{22}$$

with λ_{ij}^{ab} antisymmetric in (ij) and in (ab). The Λ equations are the stationarity conditions of the Lagrangian

$$\mathcal{L}(t,\lambda) = E_c(t) + \sum_{ia} \lambda_i^a \, R_i^a(t) + \tfrac{1}{4} \sum_{ijab} \lambda_{ij}^{ab} \, R_{ij}^{ab}(t), \qquad R_\mu(t) = \langle \Phi_\mu | \bar{H} | 0 \rangle = n_\mu(t) - D_\mu t_\mu, \tag{23}$$

where n_μ is the numerator of Section 4 ($t_\mu = n_\mu / D_\mu$ is the amplitude equation). $\mathcal{L}$ is holomorphic in t and λ (neither is ever complex conjugated), so $\partial\mathcal{L}/\partial t_\mu = 0$ gives *linear* equations for λ with the same complex integrals as the T equations; at convergence $\mathcal{L} = E_c$. With complex orbitals the only change with respect to the real-orbital formulas is again which member of a conjugate pair of integrals appears. The inhomogeneous terms are f_{ia} and $\langle ij\|ab\rangle$ with the occupied indices in the bra and *not* conjugated (the complex conjugates of the T-equation driving terms); the conjugated integrals $\overline{\langle jk\|ic\rangle} = \langle ic\|jk\rangle$ and $\overline{\langle ja\|cb\rangle} = \langle bc\|aj\rangle$ enter only through W^{ovoo} and W^{vvvo} below. The equations are those of PySCF's `gccsd_lambda.py` (`make_intermediates`, `update_lambda`) in the conventions of this note, written out from the same data that was evaluated with `numpy` and compared with PySCF (Section 9.8).

9.1 T-dependent intermediates

$\tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$ as in Section 4. The intermediate F_{kc} below equals F_{me} of Section 4.2. $\mathcal{A}, \mathcal{B}$ are one-body, $\mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{G}$ and the four W two-body building blocks; all are independent of λ and built once.

$$\begin{aligned} \mathcal{A}_{ba} = f_{ba} &- \sum_j f_{ja} \, t_j^b \\ &- \sum_{jc} \langle jb\|ac\rangle \, t_j^c + \tfrac{1}{2} \sum_{jkc} \langle jk\|ca\rangle \, \tau_{jk}^{bc} \end{aligned}$$

$$\mathcal{B}_{ij} = f_{ij} + \sum_{b} f_{ib}\, t_j^b - \sum_{kb} \langle ki \| jb \rangle\, t_k^b + \tfrac{1}{2} \sum_{kbc} \langle ik \| bc \rangle\, \tau_{jk}^{bc}$$

$$\mathcal{C}_{ijkl} = \sum_{cd} \langle ij \| cd \rangle\, \tau_{kl}^{cd}$$

$$\mathcal{D}_{jcbk} = \sum_{ld} \langle lj \| db \rangle\, t_{kl}^{cd} + \langle jc \| bk \rangle$$

$$F_{kc} = f_{kc} - \sum_{ld} \langle kl \| dc \rangle\, t_l^d$$

$$\mathcal{E}_{bj} = f_{bj} + \sum_{kc} f_{kc}\, t_{jk}^{bc} + \sum_{kc} F_{kc}\, t_k^b\, t_j^c - \tfrac{1}{2} \sum_{klc} \langle kl \| jc \rangle\, t_{kl}^{bc} + \tfrac{1}{2} \sum_{kcd} \langle kb \| dc \rangle\, t_{jk}^{cd}$$

$$\mathcal{G}_{ck} = \mathcal{E}_{ck} + \sum_{jb} \mathcal{D}_{jcbk}\, t_j^b + \sum_{b} \mathcal{A}_{cb}\, t_k^b - \sum_{j} \mathcal{B}_{jk}\, t_j^c$$

$$W_{ijkl}^{\mathrm{oooo}} = \tfrac{1}{2} \langle ij \| kl \rangle + \tfrac{1}{4}\, \mathcal{C}_{ijkl} + \sum_{c} \langle ij \| kc \rangle\, t_l^c$$

$$W_{jcbk}^{\mathrm{ovvo}} = \mathcal{D}_{jcbk} - \sum_{ld} \langle lj \| db \rangle\, t_l^c\, t_k^d - \sum_{l} \langle lj \| kb \rangle\, t_l^c + \sum_{d} \langle jc \| bd \rangle\, t_k^d$$

$$W_{icjk}^{\mathrm{ovoo}} = \tfrac{1}{4} \sum_{bd} \langle ic \| db \rangle\, \tau_{jk}^{db} + \tfrac{1}{2}\, \overline{\langle jk \| ic \rangle} + \sum_{b} \mathcal{D}_{icbk}\, t_j^b - \sum_{lb} \langle li \| jb \rangle\, t_{kl}^{cb}$$

$$W_{bcak}^{\mathrm{vvvo}} = \sum_{j} \mathcal{D}_{jcak}\, t_j^b + \tfrac{1}{4} \sum_{jl} \langle jl \| ka \rangle\, \tau_{jl}^{bc} - \tfrac{1}{2}\, \overline{\langle ka \| cb \rangle} + \sum_{jd} \langle jb \| ad \rangle\, t_{kj}^{cd}$$

9.2 Λ update

The λ-dependent intermediates are

$$M^{\mathrm{v}}_{ba} = \tfrac{1}{2}\sum_{klc} \lambda^{ca}_{kl}\, t^{cb}_{kl}$$

$$M^{\mathrm{o}}_{ij} = \tfrac{1}{2}\sum_{kcd} \lambda^{cd}_{ki}\, t^{cd}_{kj}$$

$$N^{\mathrm{v}}_{ba} = M^{\mathrm{v}}_{ba} + \sum_{k} \lambda^{a}_{k}\, t^{b}_{k}$$

$$N^{\mathrm{o}}_{ik} = M^{\mathrm{o}}_{ik} + \sum_{c} \lambda^{c}_{i}\, t^{c}_{k}$$

$$S_{ijkl} = \sum_{cd} \lambda^{cd}_{ij}\, \tau^{cd}_{kl}$$

$$U_{ijck} = \sum_{d} \lambda^{cd}_{ij}\, t^{d}_{k}$$

$$\begin{aligned} M_{ijab} = {} & \sum_{kl} \lambda^{ab}_{kl}\, W^{\mathrm{oooo}}_{ijkl} + \tfrac{1}{4}\sum_{kl} \langle kl\|ab\rangle\, S_{ijkl} \\ & - \sum_{kc} \langle kc\|ba\rangle\, U_{ijck} + \tfrac{1}{2}\sum_{cd} \lambda^{cd}_{ij}\, \langle cd\|ab\rangle \end{aligned}$$

$$\begin{aligned} q_{jb} = {} & t^{b}_{j} + \sum_{kc} \lambda^{c}_{k}\, t^{cb}_{kj} \\ & - \sum_{d} N^{\mathrm{v}}_{bd}\, t^{d}_{j} - \sum_{l} M^{\mathrm{o}}_{lj}\, t^{b}_{l} \end{aligned}$$

The term $\frac{1}{2}\sum_{cd}\lambda^{cd}_{ij}\langle cd\|ab\rangle$ in M_{ijab} is the Λ-equation particle–particle ladder: like the T-equation ladder it needs only the $vvvv$ slabs, and $\sum_c \langle kc\|ba\rangle\, U_{ijck}$ needs the $ovvv$ slabs, so the streamed scheme of Section 10 applies unchanged. The diagonal Fock elements are moved to the denominators, $\mathcal{A}'_{ba} = \mathcal{A}_{ba} - \delta_{ba} f_{aa}$ and $\mathcal{B}'_{ij} = \mathcal{B}_{ij} - \delta_{ij} f_{ii}$ (they are not part of the right-hand sides). With $P(ij)X_{ijab} = X_{ijab} - X_{jiab}$ and $P(ab)X_{ijab} = X_{ijab} - X_{ijba}$ the numerator of the doubles equation is

$$\mathcal{N}^{ab}_{ij} = \langle ij\|ab\rangle + M_{ijab} + P(ij)P(ab)\, X_{ijab} - P(ab)\, Y_{ijab} + P(ij)\, Z_{ijab}, \tag{24}$$

with

$$X_{ijab} = \lambda^{a}_{i}\, F_{jb} + \sum_{kc} \lambda^{ca}_{ki}\, W^{\mathrm{ovvo}}_{jcbk}$$

$$Y_{ijab} = \sum_k \lambda^a_k \langle ij\|kb\rangle + \sum_c \lambda^{ca}_{ij}\, \mathcal{A}'_{cb} + \sum_c N^{\mathrm{v}}_{ca} \langle ij\|cb\rangle$$

$$Z_{ijab} = \sum_c \lambda^c_j \langle ic\|ab\rangle + \sum_k \lambda^{ab}_{ki}\, \mathcal{B}'_{jk} - \sum_k N^{\mathrm{o}}_{ik} \langle kj\|ab\rangle$$

and that of the singles equation

$$\begin{aligned}\mathcal{N}^a_i = f_{ia} &+ \sum_{jb} \lambda^b_j \langle ib\|aj\rangle \\ &+ \sum_b \lambda^b_i\, \mathcal{A}'_{ba} - \sum_j \lambda^a_j\, \mathcal{B}'_{ij} \\ &- \sum_{jkc} \lambda^{ca}_{kj}\, W^{\mathrm{ovoo}}_{icjk} - \sum_{kbc} \lambda^{bc}_{ik}\, W^{\mathrm{vvvo}}_{bcak} \\ &+ \sum_{jb} M_{ijab}\, t^b_j + \sum_{jb} \lambda^{ba}_{ji}\, \mathcal{G}_{bj} \\ &+ \sum_{jb} \langle ji\|ba\rangle\, q_{jb} + \sum_{bc} \langle ic\|ab\rangle\, N^{\mathrm{v}}_{bc} \\ &- \sum_{jk} \langle ji\|ka\rangle\, N^{\mathrm{o}}_{kj} - \sum_k M^{\mathrm{o}}_{ik}\, F_{ka} \\ &- \sum_c M^{\mathrm{v}}_{ca}\, F_{ic}\end{aligned}$$

Both are divided by the same denominators as the T equations,

$$\lambda^a_i \leftarrow \frac{\mathcal{N}^a_i}{D^a_i}, \qquad \lambda^{ab}_{ij} \leftarrow \frac{\mathcal{N}^{ab}_{ij}}{D^{ab}_{ij}}, \qquad D^a_i = f_{ii} - f_{aa}, \quad D^{ab}_{ij} = f_{ii} + f_{jj} - f_{aa} - f_{bb}, \tag{25}$$

iterated from $\lambda = t$ with DIIS and an optional level shift exactly like the amplitudes. The cost per iteration is that of one T iteration (the same contractions, with λ replacing t in the ladder).

9.3 One-particle density matrix

Let $\gamma_{pq} = \langle 0|(1+\Lambda)e^{-T} a^\dagger_p a_q e^T|0\rangle$, so that the one-body energy is $\sum_{pq} h_{pq}\gamma_{pq}$ (orbital relaxation is not included). In the occupied–virtual partitioning $\gamma_{pq} = \delta_{pq} n_p + D_{qp}$, with the blocks D^{oo}_{ij}, D^{vv}_{ab}, D^{vo}_{ai} and $D^{\mathrm{ov}}_{ia} = \lambda^a_i$ given by

$$D^{\mathrm{oo}}_{ij} = -\sum_e \lambda^e_i\, t^e_j - \tfrac{1}{2} \sum_{mef} \lambda^{ef}_{im}\, t^{ef}_{jm}$$

$$D^{\mathrm{vv}}_{ab} = \sum_m t^a_m\, \lambda^b_m + \tfrac{1}{2} \sum_{mne} t^{ea}_{mn}\, \lambda^{eb}_{mn}$$

$$x^{(1)}_{mi} = \tfrac{1}{2} \sum_{nef} \lambda^{ef}_{mn}\, t^{ef}_{in}$$

$$x^{(2)}_{ae} = \tfrac{1}{2} \sum_{mnf} t^{fa}_{mn}\, \lambda^{fe}_{mn} + \sum_{m} t^{a}_{m}\, \lambda^{e}_{m}$$

$$D^{\mathrm{vo}}_{ai} = \sum_{me} t^{ae}_{im}\, \lambda^{e}_{m} - \sum_{m} x^{(1)}_{mi}\, t^{a}_{m} - \sum_{e} t^{e}_{i}\, x^{(2)}_{ae} + t^{a}_{i}$$

The coupled-cluster (Lagrangian) density is not Hermitian, already for real orbitals (e.g. D^{ov} and D^{vo} differ); the Hermitian part, which is what the real part of the energy and of its derivatives involves, is

$$\bar{\gamma}_{pq} = \tfrac{1}{2}\big(\gamma_{pq} + \gamma^{*}_{qp}\big), \qquad \mathrm{Re}\sum_{pq} h_{pq}\gamma_{pq} = \sum_{pq} h_{pq}\bar{\gamma}_{pq}. \tag{26}$$

PySCF returns $\bar{\gamma}^{\mathsf{T}}$ (its `dm1[p,q]` is $\bar{\gamma}_{qp} = \langle a^{\dagger}_{q} a_{p}\rangle$).

9.4 Two-particle density matrix

Let $\Gamma_{pqrs} = \langle 0|(1+\Lambda)e^{-T} a^{\dagger}_{p} a^{\dagger}_{q} a_{s} a_{r} e^{T}|0\rangle$, so the energy is

$$E = \sum_{pq} h_{pq}\bar{\gamma}_{pq} + \tfrac{1}{4}\sum_{pqrs} \langle pq\|rs\rangle\, \Gamma_{pqrs} \quad \text{(real part)}, \tag{27}$$

and Γ is antisymmetric in (pq) and in (rs), with $\Gamma^{*}_{pqrs} = \Gamma_{rspq}$ and $\sum_{p}\Gamma_{pqps} = (N-1)\bar{\gamma}_{qs}$. It is assembled in four steps; the notation is that of `gccsd_rdm.py`. In this part (and only here) the intermediate $\bar{\tau}^{ab}_{ij} = t^{ab}_{ij} + 2t^{a}_{i}t^{b}_{j}$ is *not* antisymmetrized: the antisymmetrization is carried out by the d-blocks below, and using the antisymmetrized τ in these formulas changes the result.

Intermediates.

$$\bar{\tau}^{ab}_{ij} = t^{ab}_{ij} + 2\, t^{a}_{i}\, t^{b}_{j}$$

$$m_{iajb} = \sum_{kc} \lambda^{ac}_{ik}\, t^{cb}_{kj}$$

$$u^{(1)}_{ia} = \sum_{kc} \lambda^{c}_{k}\, t^{ca}_{ki}$$

$$u^{(2)}_{cb} = \sum_{k} \lambda^{c}_{k}\, t^{b}_{k}$$

$$u^{(3)}_{kj} = \sum_{c} \lambda^{c}_{k}\, t^{c}_{j}$$

$$u^{(4)}_{lj} = \sum_{d} m_{ldjd}$$

$$u^{(5)}_{db} = \sum_{l} m_{ldlb}$$

$$u^{(6)}_{ijkl} = \tfrac{1}{16} \sum_{cd} \lambda^{cd}_{kl}\, \bar{\tau}^{cd}_{ij}$$

$$u^{(7)}_{ij} = \tfrac{1}{4} \sum_{c} m_{icjc}$$

$$u^{(8)}_{ab} = \tfrac{1}{4} \sum_{k} m_{kakb}$$

***g*-blocks.** The *oovv*, *ovvv* and *ooov* blocks are complex conjugated before the last (unconjugated) term of the last two is added:

$$\begin{aligned} g^{\mathrm{oovv}}_{ijab} = \Big[& \tfrac{1}{4}\, \overline{\lambda^{ab}_{ij}} + \tfrac{1}{4}\, \bar{\tau}^{ab}_{ij} \\ & + u^{(1)}_{ia}\, t^{b}_{j} + \tfrac{1}{2} \sum_{c} u^{(2)}_{cb}\, t^{ca}_{ij} \\ & + \tfrac{1}{2} \sum_{k} \bar{\tau}^{ab}_{ki}\, u^{(3)}_{kj} - \tfrac{1}{4} \sum_{l} u^{(4)}_{lj}\, \bar{\tau}^{ba}_{li} \\ & - \tfrac{1}{4} \sum_{d} u^{(5)}_{db}\, \bar{\tau}^{da}_{ji} - \tfrac{1}{2} \sum_{ld} m_{ldia}\, \bar{\tau}^{bd}_{lj} \\ & + \sum_{kl} u^{(6)}_{ijkl}\, \bar{\tau}^{ab}_{kl} \Big]^{*} \end{aligned}$$

$$g^{\mathrm{vvvv}}_{abcd} = \tfrac{1}{8} \sum_{ij} \bar{\tau}^{ab}_{ij}\, \lambda^{cd}_{ij}$$

$$g^{\mathrm{oooo}}_{klij} = \tfrac{1}{8} \sum_{ab} \lambda^{ab}_{kl}\, \bar{\tau}^{ab}_{ij}$$

$$\begin{aligned} g^{\mathrm{ooov}}_{jkia} = \Big[& - \tfrac{1}{4} \sum_{b} \bar{\tau}^{ba}_{jk}\, \lambda^{b}_{i} + \sum_{l} g^{\mathrm{oooo}}_{iljk}\, t^{a}_{l} \\ & - u^{(7)}_{ij}\, t^{a}_{k} + \tfrac{1}{2} \sum_{c} m_{icja}\, t^{c}_{k} \Big]^{*} + \tfrac{1}{4} \sum_{b} \lambda^{ab}_{jk}\, t^{b}_{i} \end{aligned}$$

$$\begin{aligned} g^{\mathrm{ovvo}}_{ibaj} = {} & \lambda^{a}_{i}\, t^{b}_{j} + m_{iajb} \\ & - \sum_{kc} \lambda^{ac}_{ik}\, t^{c}_{j}\, t^{b}_{k} \end{aligned}$$

$$g^{\mathrm{ovvv}}_{iabc} = \Big[\tfrac{1}{4}\sum_j \lambda^a_j\,\bar{\tau}^{bc}_{ij} + \sum_d g^{\mathrm{vvvv}}_{bcad}\,t^d_i + u^{(8)}_{ac}\,t^b_i + \tfrac{1}{2}\sum_k m_{kaib}\,t^c_k\Big]^* + \tfrac{1}{4}\sum_j \lambda^{bc}_{ij}\,t^a_j$$

Here g^{oooo} in g^{ooov} is the unconjugated block, and the g^{ovvo} and g^{vvvv} blocks enter the d-blocks without conjugation.

d-blocks. Antisymmetrizing and symmetrizing the g-blocks,

$$\begin{aligned}
d^{\mathrm{ovov}}_{iajb} &= g^{\mathrm{oovv}}_{ijab} - g^{\mathrm{oovv}}_{ijba}\\
d^{\mathrm{vvvv}}_{abcd} &= g^{\mathrm{vvvv}}_{acbd} - g^{\mathrm{vvvv}}_{acdb}\\
d^{\mathrm{oooo}}_{ijkl} &= g^{\mathrm{oooo}}_{ikjl} - g^{\mathrm{oooo}}_{iklj}\\
d^{\mathrm{ovvv}}_{iabc} &= g^{\mathrm{ovvv}}_{ibac} - g^{\mathrm{ovvv}}_{ibca}\\
d^{\mathrm{ooov}}_{ijka} &= g^{\mathrm{ooov}}_{ikja} - g^{\mathrm{ooov}}_{kija}\\
d^{\mathrm{ovvo}}_{iabj} &= g^{\mathrm{ovvo}}_{ibaj}
\end{aligned}$$

and then

$$\begin{aligned}
&d^{\mathrm{ovov}}_{iajb} \leftarrow \tfrac{1}{2}\big(d^{\mathrm{ovov}}_{iajb} + d^{\mathrm{ovov}}_{jbia}\big),\\
&d^{\mathrm{vvvv}}_{abcd} \leftarrow d^{\mathrm{vvvv}}_{abcd} + \overline{d^{\mathrm{vvvv}}_{badc}}, \qquad d^{\mathrm{oooo}}_{ijkl} \leftarrow d^{\mathrm{oooo}}_{ijkl} + \overline{d^{\mathrm{oooo}}_{jilk}},\\
&d^{\mathrm{ovvo}}_{iabj} \leftarrow \tfrac{1}{2}\big(d^{\mathrm{ovvo}}_{iabj} + \overline{d^{\mathrm{ovvo}}_{jbai}}\big).
\end{aligned}$$

Assembly. Define the array $\mathcal{R}_{pqrs} \equiv \Gamma_{qspr}$ blockwise (i, j, k occupied; a, b, c virtual),

$$\begin{aligned}
\mathcal{R}_{iajb} &= d^{\mathrm{ovov}}_{iajb}\\
\mathcal{R}_{aibj} &= \overline{d^{\mathrm{ovov}}_{iajb}}\\
\mathcal{R}_{ijab} &= -d^{\mathrm{ovvo}}_{ibaj}\\
\mathcal{R}_{abij} &= -d^{\mathrm{ovvo}}_{ibaj}\\
\mathcal{R}_{iabj} &= d^{\mathrm{ovvo}}_{iabj}\\
\mathcal{R}_{aijb} &= \overline{d^{\mathrm{ovvo}}_{iabj}}\\
\mathcal{R}_{abcd} &= d^{\mathrm{vvvv}}_{abcd}\\
\mathcal{R}_{ijkl} &= d^{\mathrm{oooo}}_{ijkl}\\
\mathcal{R}_{iabc} &= d^{\mathrm{ovvv}}_{iabc}\\
\mathcal{R}_{abic} &= d^{\mathrm{ovvv}}_{icab}\\
\mathcal{R}_{abci} &= \overline{d^{\mathrm{ovvv}}_{icba}}\\
\mathcal{R}_{aibc} &= \overline{d^{\mathrm{ovvv}}_{iacb}}\\
\mathcal{R}_{ijka} &= d^{\mathrm{ooov}}_{ijka}\\
\mathcal{R}_{iajk} &= d^{\mathrm{ooov}}_{jkia}\\
\mathcal{R}_{ijak} &= \overline{d^{\mathrm{ooov}}_{jika}}\\
\mathcal{R}_{aijk} &= \overline{d^{\mathrm{ooov}}_{kjia}}
\end{aligned}$$

where the block of $\mathcal{R}$ is selected by the occupied/virtual character of p, q, r, s in that order. The correlation part of the one-particle matrix, $\bar{\delta}_{pq} \equiv \bar{\gamma}_{qp} - \delta_{pq} n_p$ (with $n_p = 1$ for occupied p), is then added for every occupied i,

$$\mathcal{R}_{ii\,pq} \mathrel{+}= \bar{\delta}_{pq}, \quad \mathcal{R}_{pq\,ii} \mathrel{+}= \bar{\delta}_{pq}, \quad \mathcal{R}_{p\,ii\,q} \mathrel{-}= \bar{\delta}_{pq}, \quad \mathcal{R}_{i\,pq\,i} \mathrel{-}= \bar{\delta}_{qp}, \tag{28}$$

followed by the mean-field part, $\mathcal{R}_{iijj} \mathrel{+}= 1$ and $\mathcal{R}_{ijji} \mathrel{-}= 1$ for all occupied i, j, and finally

$$\Gamma_{pqrs} = \mathcal{R}_{rpsq}. \tag{29}$$

(For real orbitals every conjugation is trivial and Γ reduces to the usual 2-RDM of spin-orbital CCSD.)

9.5 Storage

Of the g/d-blocks only g^{vvvv} is $O(v^4)$. To avoid ever storing it, contract it directly with the integrals: $\frac{1}{4}\sum\langle ab\|cd\rangle\,\Gamma_{abcd}$ with $\Gamma_{abcd} = d^{\mathrm{vvvv}}_{cadb}$ and $g^{\mathrm{vvvv}}_{abcd} = \frac{1}{8}\sum_{ij} \bar{\tau}^{ab}_{ij}\lambda^{cd}_{ij}$ is evaluated slab by slab exactly like the T ladder, $O(o^2v^4)$ time and no v^4 memory. Any other use of Γ_{abcd} needs it explicitly.

9.6 Density matrix files (`CC_RDM`)

`src/CC/lambda_Uccsd.{h,cpp}` solves the Λ equations above (Jacobi sweeps with DIIS from $\lambda = t$, the same convergence criterion as T; the $vvvv$ ladder is streamed one slab $\langle c\,d|a\,b\rangle$ at a time, the $ovvv$ block $\langle ma\|ef\rangle$ is held once) and writes

- `<input>_MO.dm1`: records (`int32` p, q; `float64` Re, Im) with $\bar{\gamma}_{pq} = \langle a^\dagger_p a_q\rangle$ (Hermitian part), all p, q;
- `<input>_MO.dm2`: records (`int32` p, q, r, s; `float64` Re, Im) with $\Gamma_{pqrs} = \langle a^\dagger_p a^\dagger_q a_s a_r\rangle$ for $p < q$, $r < s$ only (the remaining elements follow from $\Gamma_{pqrs} = -\Gamma_{qprs} = -\Gamma_{pqsr}$); the energy is $E = \sum_{pq} h_{pq}\bar{\gamma}_{pq} + \frac{1}{2}\sum_{pqrs}\langle pq|rs\rangle\Gamma_{pqrs}$;

both ended by an all-zero record. Indices are 1-based and in Kramers-pair order, $(1, 2) = (\psi_1, \bar{\psi}_1)$, $(3, 4) = (\psi_2, \bar{\psi}_2), \ldots$: X2C and the positive-energy block of the 4-component run are numbered that way already, the non-relativistic spin-orbitals are renumbered $\alpha_k \to 2k + 1$, $\beta_k \to 2k + 2$. Frozen-core spinors are included with their mean-field contributions (they appear only through the one-body and mean-field terms of Section 9.4). Elements with $|\Gamma|$ below `CC_RDM_THRESHOLD` are omitted, and the energy recomputed from the written elements is printed next to $E_0 + E_c$ as a check (the two agree to 10^{-9} Eh for the default 10^{-10}). The 2-RDM is produced element by element from the d-blocks (never as a dense n^4 array); only the all-virtual block is formed slab by slab from $\bar{\tau}$ and λ. The densities are unrelaxed: no orbital response is included.

9.7 Densities in the basis of the Gaussian primitives (`src/CC/gauss_rdm.{h,cpp}`)

With `CC_RDM_GAUSSIAN` (default) the same densities are also written over the Cartesian Gaussian *primitives* $\chi_\mu = N_\mu x^{l_x} y^{l_y} z^{l_z} e^{-\alpha_\mu|\mathbf{r}-\mathbf{A}|^2}$ in the formats read by `RHO2_OPS`: `<input>_GAU.basis`, `<input>_GAU.dm1`, `<input>_GAU.dm2`.

Expansion. Every contracted Cartesian AO is unrolled into its primitives (the libcint factor of its shell times its contraction coefficient is the weight of the plain monomial), and identical primitives (same centre, exponent and powers) are merged. A spinor has one coefficient vector per spin/large–small component c: $\varphi_p^{(c)} = \sum_\mu C^c_{\mu p}\chi_\mu$, with $c \in \{\alpha,\beta\}$ over the Large primitives for NON_REL and X2C (the NON_REL spin-orbitals are pure α or β, the spherical transformation and the `LOWGEN` reduction are folded into C), and $c \in \{L\alpha, L\beta, S\alpha, S\beta\}$ for the 4-component spinors, the Small component being expanded over the raw RKB small AOs (their own primitives, kept distinct from the Large ones). The weights are divided by $N_\mu = (2\alpha/\pi)^{3/4}(4\alpha)^{l/2}/\sqrt{(2l_x-1)!!(2l_y-1)!!(2l_z-1)!!}$ (the normalization `RHO2_OPS` multiplies by); `CC_RDM_CNORM1 TRUE` keeps it inside D instead (`$CNORM1`).

Definition. The spin- and component-summed pair density is $\rho_2(\mathbf{r}_1,\mathbf{r}_2) = \sum D_{\mu\nu\lambda\sigma}\,\chi_\mu(\mathbf{r}_1)\chi_\lambda(\mathbf{r}_1)\chi_\nu(\mathbf{r}_2)\chi_\sigma(\mathbf{r}_2)$,

$$D^{\mathrm{raw}}_{\mu\nu\lambda\sigma} = \sum_{c_1c_2}\sum_{pqrs}\Gamma_{pqrs}\,\overline{C^{c_1}_{\mu p}}\,C^{c_1}_{\lambda r}\,\overline{C^{c_2}_{\nu q}}\,C^{c_2}_{\sigma s}, \tag{30}$$

so only pairs of primitives of the same component occur (μ,λ both Large or both Small, and likewise ν,σ), and $\sum D_{\mu\nu\lambda\sigma}S_{\mu\lambda}S_{\nu\sigma} = N(N-1)$. The real basis functions make $\chi_\mu\chi_\lambda$ symmetric in (μ,λ) and $\chi_\nu\chi_\sigma$ in (ν,σ), and Γ is Hermitian, $D^{\mathrm{raw}}_{\lambda\sigma\mu\nu} = \overline{D^{\mathrm{raw}}_{\mu\nu\lambda\sigma}}$, so the average over the four permutations $D^S = \frac{1}{4}\sum_{\mathrm{perm}} D^{\mathrm{raw}} = \frac{1}{2}\,\mathrm{Re}\left(D^{\mathrm{raw}}_{\mu\nu\lambda\sigma} + D^{\mathrm{raw}}_{\lambda\nu\mu\sigma}\right)$ is exactly real: the imaginary parts cancel by complex conjugation, and only D^S is written (the largest imaginary part of the symmetrized element is printed as a check). The 1-RDM is $\gamma^S_{\mu\lambda} = \frac{1}{2}\,\mathrm{Re}\left(\gamma_{\mu\lambda} + \gamma_{\lambda\mu}\right)$, $\gamma_{\mu\lambda} = \sum_c \sum_{pr} \overline{C^c_{\mu p}}\,\bar{\gamma}_{pr}C^c_{\lambda r}$.

Files. `.basis`: a header line, then one line `Z x y z exponent nx ny nz` per primitive, Large primitives first. `.dm2`: Fortran-unformatted-style records (`int32` record length, four `int32` indices p,q,r,s, `float64` value, `int32` record length), 1-based primitive indices, terminated by a record of four zero indices. Both D_{pqrs} and its electron-exchange partner D_{qpsr} are present, with $r \le p$ and $s \le q$ (the `RHO2_OPS` convention without `$SYMRED`); the stored value is the sum of the symmetry-equivalent elements, $m_{pr}m_{qs}D^S_{pqrs}$ with $m = 2$ when the two indices differ and 1 otherwise. `.dm1`: records (`int32` length, `int32` μ, `int32` λ, `float64` $\gamma^S_{\mu\lambda}$, `int32` length) for $\mu \ge \lambda$, terminated by $\mu = \lambda = 0$.

Algorithm. Γ is never held as an n^4 array: it is generated one slab $\Gamma_{p\cdots}$ at a time from the d-blocks. For a block of rows μ of one sector, a sweep over p accumulates $R^c_{\mu qrs} = \sum_p \overline{C^c_{\mu p}}\Gamma_{pqrs}$ and $W^c_{\mu pqs} = \sum_r C^c_{\mu r}\Gamma_{pqrs}$ (`zgemm`); then, for each μ, $Y_{\mu\lambda;qs} = \frac{1}{2}\sum_c\left(\sum_r C^c_{\lambda r}R^c_{\mu qrs} + \sum_p \overline{C^c_{\lambda p}}W^c_{\mu pqs}\right)$ for $\lambda \le \mu$ (the two-term average already is the permutation average over $\mu \leftrightarrow \lambda$), and finally $E_{\mu\lambda;\nu\sigma} = \sum_{c_2}\sum_{qs}\overline{C^{c_2}_{\nu q}}Y_{\mu\lambda;qs}C^{c_2}_{\sigma s}$, whose real part is D^S. Only $\nu \le \mu$ is computed and the electron-exchange partner is written alongside; each record is written to disk at once, so the memory is $2N_cBn^3$ complex numbers for a block of B rows (`CC_RDM_MAXMEM` sets B; more rows per block means fewer sweeps over Γ). The cost is about $4P^4n$ flops for P primitives and n spinors, plus the sweeps over Γ.

Checks and validation. Printed after each run: $\sum \gamma^S S = N$ and $\sum D^S SS = N(N-1)$ with the primitive overlaps. With `RERDMFT_CC_RDM_GAU_CHECK=1`, $V_{ee} = \frac{1}{2}\sum D\,(\mu\lambda|\nu\sigma)$ is also recomputed with the integrals of the primitives themselves (small systems): LiH/6-31G (NON_REL, X2C and 4-component, with the Small primitives) and water/cc-pVDZ with a frozen core agree with the molecular-basis two-body energy to 10^{-11}–10^{-13} Eh with dense integrals (with RI the difference

is the RI error). `RHO2_OPS` reads the files directly and reports $\mathrm{Tr}[^2D] = N(N-1)/2$ (for LiH, V_{ee} from the intracule converges to the molecular value as its radial/angular quadratures are refined).

9.8 Numerical check

PySCF 2.14 (`GCCSD`) is the reference. All checks use complex integrals: a Hartree–Fock reference of LiH is rotated inside the occupied and virtual spaces by random unitaries e^A, $A = -A^\dagger$ complex (and, in a second set, with a small complex occupied–virtual mixing so that $f_{ov} \neq 0$ and E_c acquires a small imaginary part).

- The equations above, evaluated from the same data, reproduce PySCF's intermediates and one full Λ update to 10^{-15}, and the d-blocks and $\mathcal{R}$ array to 10^{-16}.
- $\partial\mathcal{L}/\partial t_\mu$ at PySCF's converged (t, λ) vanishes to 3×10^{-12} for all singles and all antisymmetry-unique doubles.
- Central finite differences of $\mathrm{Re}\, E_{\mathrm{CC}}$ (amplitudes re-solved) with respect to a Hermitian perturbation of h and of the antisymmetrized integrals equal $\sum h'_{pq}\bar{\gamma}_{pq}$ and $\frac{1}{4}\sum\langle pq\|rs\rangle'\Gamma_{pqrs}$ to 10^{-8}.
- $\sum h\bar{\gamma} + \frac{1}{4}\sum\langle pq\|rs\rangle\Gamma_{pqrs} = \mathrm{Re}(E_0 + E_c)$ to 10^{-14}.

10 Implementation (`src/CC/Uccsd.{h,cpp}`)

The code is a transcription of Section 4 with two reorganizations that do not change the equations. Every contraction is a two-operand einsum evaluated as one `zgemm`.

- **No stored $\langle ma\|ef\rangle$ or $\langle ab\|ef\rangle$ blocks.** The integrals are supplied as a Coulomb factorization $(pq|rs) = \sum_P \mathrm{bra}_P(p,q)\,\mathrm{ket}_P(r,s)$ (RI: bra = ket = B_P; Cholesky: $\mathrm{bra}_L(a,c) = V_L(c,a)$, $\mathrm{ket}_L(b,d) = V_L^*(b,d)$), and one slab with a fixed first index, $\langle p\,a|ef\rangle$, is rebuilt by a single `zgemm` whenever it is needed. A loop over occupied m accumulates every term containing $\langle ma\|ef\rangle$ (F_{ae}, the T_1 equation, W_{mbej}, the $P(ij)\,t_i^e\langle ab\|ej\rangle$ term and the intermediate Z below); a loop over virtual a applies the particle–particle ladder. Memory is $O(o^2v^2 + N_{\mathrm{aux}}n^2)$, plus $O(N_{\mathrm{aux}}v^2)$ for the cached second factor. For dense integrals the same slabs are read element by element.
- **The ladder is applied as $\sum_{ef}\tau_{ij}^{ef}\langle ab|ef\rangle$.** Because τ is antisymmetric in (ef), $\frac{1}{2}\sum_{ef}\tau_{ij}^{ef}\langle ab\|ef\rangle = \sum_{ef}\tau_{ij}^{ef}\langle ab|ef\rangle$, so no antisymmetrized $vvvv$ slab is formed. The two other pieces of $\frac{1}{2}\sum\tau W_{abef}$ are $-P(ab)\sum_m t_m^b Z_{ijm}^a$ with $Z_{ijm}^a = \frac{1}{2}\sum_{ef}\tau_{ij}^{ef}\langle am\|ef\rangle$, and the $\tau\tau$ term, which is moved into W_{mnij} by doubling its coefficient there ($\frac{1}{4}\to\frac{1}{2}$): $\frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{2}\cdot\frac{1}{2}$.

Amplitudes are updated by Jacobi sweeps, $t \leftarrow (\mathrm{RHS} - \sigma t)/(D - \sigma)$ with level shift σ, accelerated by DIIS on the antisymmetry-unique amplitudes (real weights, `Utils/DIIS.h`). The Fock matrix is built from the integrals (frozen occupied spinors included), so $f_{ov} \neq 0$ and non-canonical orbitals are handled. Checked against PySCF 2.14: LiH/6-31G, water/cc-pVDZ with frozen core and CO/cc-pVDZ with frozen core (dense and Cholesky agree to 10^{-10}–10^{-8} Eh with PySCF's RCCSD), and the $c \to \infty$ limit of the X2C and 4-component references reproduces the non-relativistic result.